

Finding mathematical patterns with data science

Complete chapter candidate

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1 Finding mathematical patterns with data science

The normalized sum of symplectic ridge areas is a useful coarse indicator of systolic ratio, but does not reliably order the bodies selected for small ridge sums. Its strong inverse association in the broad sample weakens markedly after restriction to either low ridge sums or high ratios. Fresh low-ridge selections still have larger mean ratios than comparison bodies, even where stronger filtering gives no further gain. The distinction is between identifying a promising population and finding an improving direction within it.

Affine shape variation of the polygon factors contributes materially to this pattern. Balancing their vertex covariances raises the mean numerical ratio from 0.342 to 0.654 in a matched sample of 320 products, while the pooled correlation of within-group ranks weakens from -0.929 to -0.249 . This supplies a concrete geometric source of variation in the broad sample. It does not explain all of the remaining association or turn balancing into a universal improvement rule.

Exact geometry imposes complementary limits on any explanation. Among polygon products the normalized ridge sum is at least eight, and the cube attains that minimum with systolic ratio only $1/2$. Minimizing ridge area therefore cannot maximize the ratio universally. Yet rotated regular pentagon products satisfy an exact inverse-square relation between the two quantities. A broader account must accommodate both this strict family relation and its failure outside that family.

1.1 The geometric quantities and the sampled populations

For numerical evaluations we write $\widehat{\text{sys}} = c_{\text{num}}^2/(2V_{\text{num}})$. Its normalization agrees with sys ; its volume is computed in floating-point arithmetic. The empirical results below concern this numerical ratio.

A two-dimensional face of a four-dimensional polytope is called a ridge. For a ridge F with cyclically ordered vertices $v_0, \dots, v_{r-1}$, define its absolute symplectic area by

$$A_\omega(F) = \frac{1}{2} \left| \sum_{i=0}^{r-1} \omega_0(v_i, v_{i+1}) \right|, \quad v_r = v_0, \quad \omega_0 = dq_1 \wedge dp_1 + dq_2 \wedge dp_2.$$

This is the polygonal area formula with the symplectic pairing in place of the planar determinant. It measures the area seen by ω_0 ; a ridge can have positive Euclidean area and zero symplectic area. Two summaries proved useful:

$$R(K) = \frac{\sum_F A_\omega(F)}{\sqrt{\text{vol}_4(K)}}, \quad C(K) = \frac{\max_F A_\omega(F)}{\sum_F A_\omega(F)}.$$

The first records total symplectic ridge area relative to volume. The second records the fraction concentrated on the largest ridge, when the total is positive. Both are unchanged by translation, common dilation and linear symplectic maps. The area formula is translation invariant because the terms involving the translation vector cancel around the closed polygon.

The exploratory sample contains 4,096 generic four-dimensional polytopes, with 512 at each facet count $F = 5, \dots, 12$, and 10,240 Lagrangian products $P \times Q$, with 1,024 at each pair $3 \leq k \leq m \leq 6$ of factor side counts. In coordinates (q_1, q_2, p_1, p_2) , P lies in the q -plane and Q in the p -plane. A fixed facet count or pair of side counts defines a sample group.

The unit facet normals are independent and uniform on the appropriate sphere, and the support heights are independent and uniform in $[0.8, 1.2)$. Boundedness and the presence of every proposed facet are required. Thus these are specified distributions of bodies; the precise admission rules are in Appendix A.1. The largest numerical ratio in the exploratory sample is 0.863. Its retained values and current reevaluation are described in Appendix A.2.

1.2 A strong bulk association weakens under selection

The pooled rank correlation between R and the numerical systolic ratio is about -0.94 . Rank correlation compares the orderings of the bodies by the two quantities; a value of -1 would mean exactly reversed orderings. The ordinary linear correlation is much weaker, about -0.21 . Thus the principal observation concerns which bodies have larger systolic ratio, rather than how much the systolic ratio changes per unit of ridge area. The inverse association also occurs within every sampled facet-count or polygon-size group: the rank correlations range from about -0.84 to -1.00 . Thus it is not solely a consequence of pooling these groups. Figures 2 and 3 show the separate populations.

Conditioning the original sample on either low R or high numerical systolic ratio weakens this association. If the lowest ten percent by R are retained within each group, the pooled correlation of within-group ranks is -0.222 ; retaining the highest ten percent by ratio gives -0.286 , compared with -0.944 before restriction. Figure 1 shows the progressive attenuation. At the five-percent cutoff the high-ratio panel still has a negative association, so attenuation should not be confused with independence. The smallest panels contain only six generic or eleven product bodies per group.

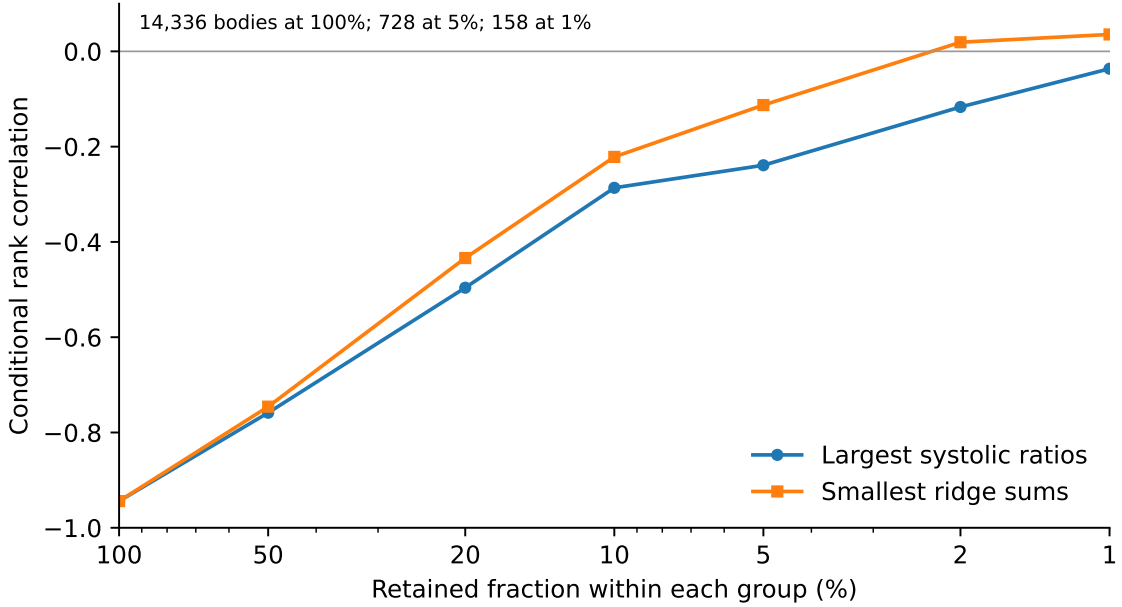


Figure 1: Restricting the original sample by either ridge sum or systolic ratio weakens their inverse association. Within each facet-count or factor-side-count group, retain the indicated fraction, rerank both quantities within that retained group, and compute the pooled Spearman correlation of the group-relative ranks. This is a descriptive conditional analysis of the original data. It does not evaluate a fresh selection rule or identify a geometric cause.

The attenuation is substantially stronger than in a Gaussian reference fitted separately to each group’s full-sample rank correlation. At the five-percent cutoff, that reference gives mean correlation -0.695 for either selection, versus the observed -0.239 for high ratios and -0.113 for low R . This reference does not reproduce the conditional pattern; it does not exclude range-restriction effects under other dependence structures. Its construction is in Appendix A.4.

Ridge measurements also contain predictive information beyond face counts and incidences. Under a common training/test split, ridge-area features alone gave test $R^2 = 0.887$, whereas combinatorial counts alone gave 0.043 . Here R^2 compares squared prediction error with prediction by the test-set mean. The feature lists, additional models and split definitions are in Appendix A.3.

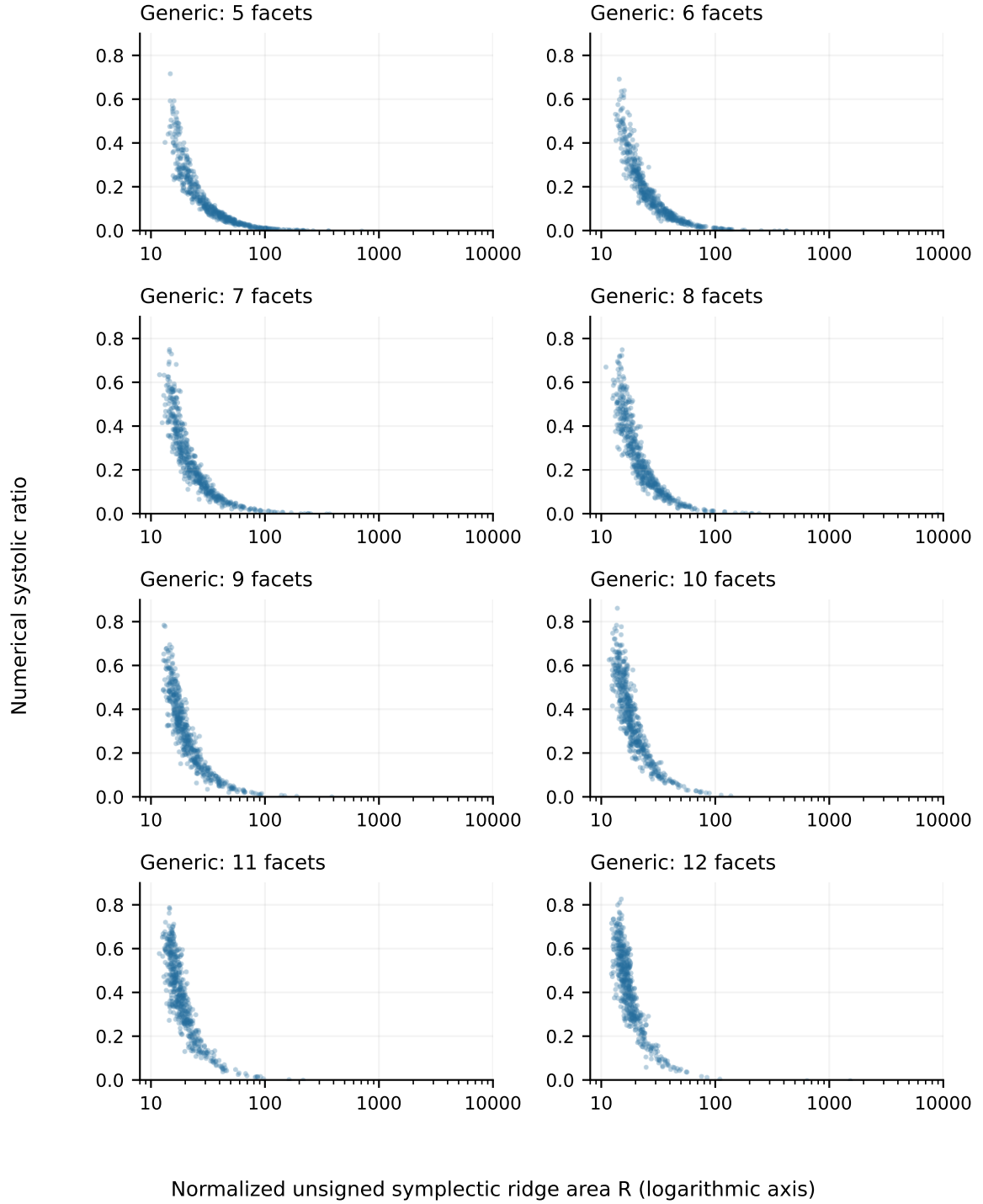


Figure 2: Ridge area and numerical systolic ratio for the 4,096 generic polytopes, separated by facet count. Each point is one body; the horizontal axis is logarithmic. All original numerical values are retained (Appendix A.2).

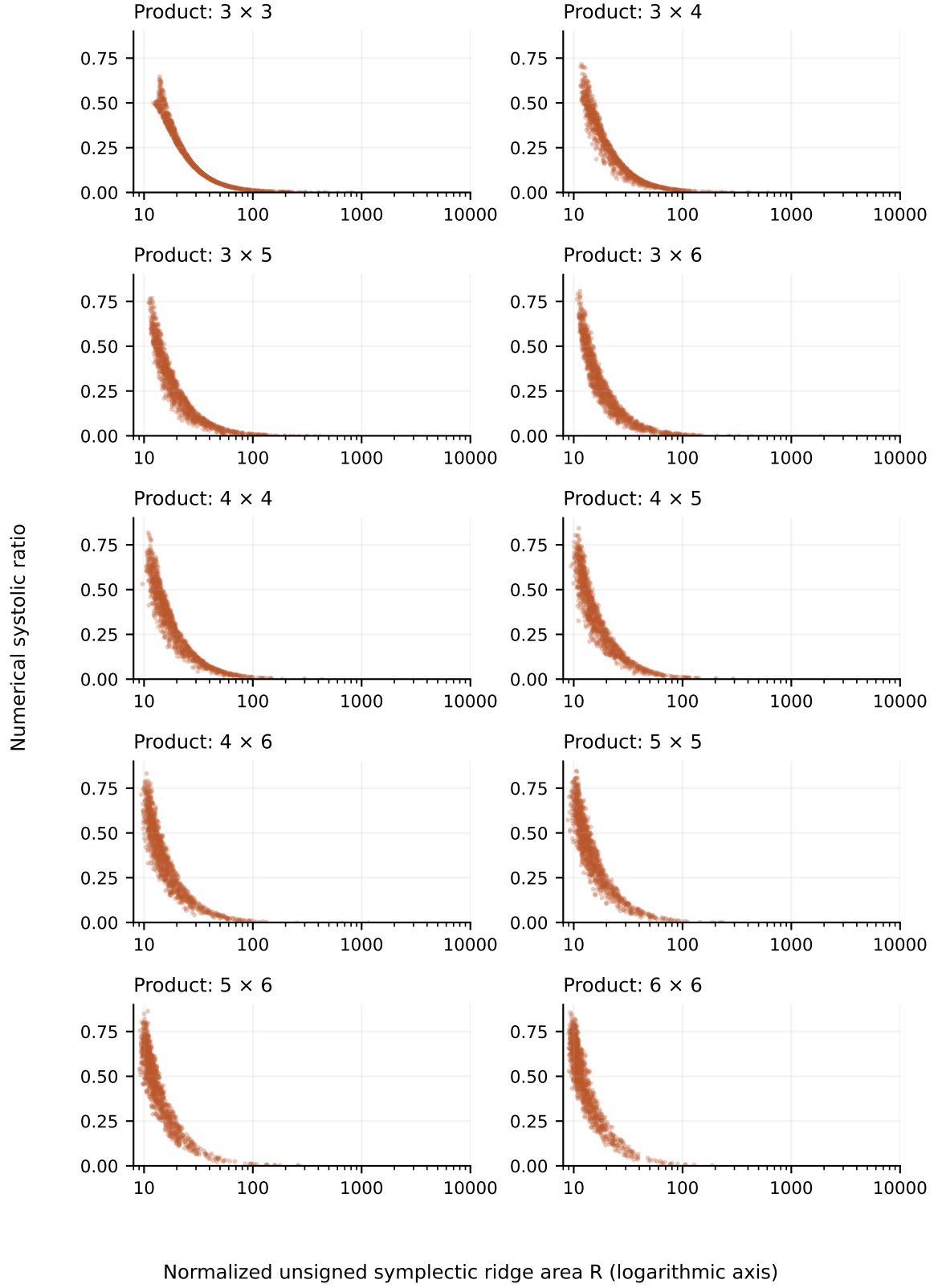


Figure 3: The same comparison for the 10,240 Lagrangian products, separated by factor side counts. Each group contains 1,024 bodies.

1.3 Exact geometry: a special law and its limits

A universal inverse relation between ridge sum and systolic ratio is false, but it can hold exactly on a restricted family. The product edge formula explains both statements and identifies the smallest possible ridge sum among polygon products.

For products the descriptor has an elementary edge formula. Let u_i and v_j be the counterclockwise boundary edge vectors of P and Q . A ridge lying entirely in a q - or p -plane has zero symplectic area. A mixed ridge is the rectangle formed from one edge in each factor, and its symplectic area is $|u_i \cdot v_j|$. Consequently

$$R(P \times Q) = \frac{\sum_{i,j} |u_i \cdot v_j|}{\sqrt{\text{area}(P) \text{area}(Q)}}.$$

For a fixed u_i , the sum over the edges of Q is the total variation of the projection of its boundary onto the direction of u_i . Convexity gives one increasing and one decreasing chain between the extreme projections, so

$$\sum_j |u_i \cdot v_j| = 2\|u_i\| \text{width}_Q(u_i/\|u_i\|).$$

The descriptor therefore combines the edge lengths of one factor with the widths of the other in the edge directions. This gives concrete geometry to investigate instead of treating the successful column as an unexplained score.

The minimum ridge sum for polygon products. Let P, Q be convex polygons with nonempty interior, set $D = P - P$, and write A_P, A_Q for their areas. Let $J_2(x, y) = (-y, x)$, and normalize mixed area, for $t \geq 0$, by

$$\text{area}(K + tL) = \text{area}(K) + 2tV(K, L) + t^2 \text{area}(L).$$

Then

$$R(P \times Q) = \frac{4V(D, J_2Q)}{\sqrt{A_P A_Q}} \geq 8. \quad (1)$$

Equality holds precisely when P is centrally symmetric and J_2Q is positively homothetic to D , allowing translation.

Proof. Let h_D be the support function of D . The boundary-projection identity gives $\sum_i |u_i \cdot v_j| = 2h_D(v_j)$. The outward unit normal to the counterclockwise edge J_2v_j is $v_j/\|v_j\|$. The polygon mixed-area formula therefore gives

$$2V(D, J_2Q) = \sum_j \|v_j\| h_D(v_j/\|v_j\|) = \sum_j h_D(v_j),$$

which proves the identity. Brunn–Minkowski and its equality case [Gar02, Sections 3 and 5] imply

$$V(D, J_2Q)^2 \geq \text{area}(D)A_Q, \quad \text{area}(D) \geq 4A_P.$$

The first inequality is an equality exactly when D and J_2Q are positively homothetic. The second is an equality exactly when P is centrally symmetric. Substitution proves the bound and equality statement. $\square$

For two aligned unit squares, eight edge pairs have absolute scalar product one and the rest zero, so $R = 8$. Their product is the unit four-dimensional cube, whose capacity is one and whose systolic ratio is $1/2$.¹ HKO has the larger ridge score 8.944 and a ratio greater than

¹For the centered unit cube the dual rows are $\pm 2e_i$. In the variational formula, closure forces equal weights x_j on the two q_j rows and equal weights y_j on the two p_j rows, with $\sum_{j=1}^2 (x_j + y_j) = 1/2$. The order of the four rows in a coordinate pair gives a contribution at most $8x_j y_j$ to Q . Thus $Q \leq 8 \sum_j x_j y_j \leq 1/2$. A cyclic word on one coordinate pair with its four weights equal to $1/4$ attains $Q = 1/2$, giving $c = 1/(2Q) = 1$.

one. Thus a universal claim that a smaller ridge score forces a larger systolic ratio is false. The strong negative association in the sampled distributions needs a more specific explanation.

In particular, the cube attains the global minimum of R among polygon products without attaining a large systolic ratio. The HKO value is also not separated from ordinary sampled bodies by R : its value 8.944 exceeds the exploratory sample minimum 8.926.

For rotated regular pentagons, the descriptor does have an exact relation to capacity. Let $K_\theta = P_5 \times \text{Rot}(\theta)P_5$, with circumradius one, and let $d(\theta) = \min_{k \in \mathbb{Z}} |\theta - k\pi/5|$. Each edge has length $2 \sin(\pi/5)$ and each factor has area $5 \sin(\pi/5) \cos(\pi/5)$. The mixed-edge formula therefore gives

$$R(K_\theta) = 4 \tan(\pi/5) \sum_{k=0}^4 |\cos(\theta + 2\pi k/5)| = 16 \sin(\pi/5) \cos d(\theta).$$

For the second equality, the absolute-cosine sum is even and has period $\pi/5$. On $[-\pi/10, \pi/10]$, the terms indexed by 0, 1, 4 are nonnegative and those indexed by 2, 3 are nonpositive. Pairing opposite indices reduces the sum to $[1 + 2 \cos(2\pi/5) - 2 \cos(4\pi/5)] \cos \theta = 4 \cos(\pi/5) \cos \theta$. Combining this with the capacity profile proved in Theorem P yields

$$\text{sys}(K_\theta) R(K_\theta)^2 = 16(3 + \sqrt{5}). \quad (2)$$

Thus smaller normalized total unsigned ridge area means larger systolic ratio within this rotation family. This supplies a geometric explanation for one instance of the observed inverse association.

More precisely, R strictly decreases and sys strictly increases as d runs from 0 to $\pi/10$. Throughout this rotation family, distinct ridge sums therefore have strictly reversed systolic ratios. The periodicity and reflection symmetry prevent strict monotonicity in the unrestricted angle θ .

The exact relation is special to this family. Along sampled rotation paths of 3×6 and 4×4 products, both R and the numerical ratio decrease at the four recorded positions. These observations do not establish monotonicity between those positions. Together with the exact cube–HKO comparison, they show why a relationship visible in one family cannot be extrapolated to every deformation or product.

1.4 Coarse selection succeeds; stronger selection need not

The bulk association has a practical consequence: geometry alone can identify fresh populations with larger numerical ratios before capacity is evaluated. Within fixed factor-side-count groups, selection for low R gave mean ratio 0.626, compared with 0.316 for comparison bodies from the same source. Among products already in the lowest one percent by R , the half with lower C had a further mean advantage of 0.034 over the other half, positive in eight of ten groups. Thus the dominance of the largest ridge contains information even when the total ridge area is small. This concerns largest-ridge dominance, rather than every notion of how evenly area is distributed.

The strong ordering in the broad samples is not retained within the low- R panels themselves. Across 1,000 evaluated products in the lowest one percent by R , correlations within the ten side-count groups range from -0.315 to $+0.301$; pooling ranks computed within each group gives $+0.032$. The most extreme tenth has a larger mean ratio than the rest in only two groups. Coarse enrichment over comparison bodies and loss of the inverse ordering within the selected population are therefore both visible in the same research program.

For generic ten-facet bodies, the related score R/N_2 , where N_2 is the number of ridges, measures normalized area per ridge. Unlike products at fixed side counts, these bodies can have different ridge counts. Selecting the lowest one percent by R/N_2 gave mean ratio 0.626, against

0.339 for comparison bodies. Within that selected panel, however, the ten most extreme scores had a mean ratio 0.032 below the remaining ninety. This is a failure to obtain further gain in that sample, not an established deterioration under every stronger filter. The comparison rules and conditional resampling summaries are retained in Appendix A.4.

The area decomposition supplies a geometric view of the same limitation. For Euclidean ridge area $E(F) > 0$, write

$$A_\omega(F) = E(F)\kappa(F), \quad \kappa(F) = A_\omega(F)/E(F).$$

Here κ measures the alignment of the ridge plane with the symplectic form relative to the Euclidean metric. Neither factor is invariant under all linear symplectic maps. In the generic selection, both mean normalized Euclidean ridge area and Euclidean-area-weighted κ were smaller than in the comparison panel: 0.917 versus 1.365, and 0.377 versus 0.503. Within the selected panel, the ten lowest ridge scores instead had slightly larger ordinary area and smaller alignment factor than the other ninety, without a larger mean ratio. Lowering one component is therefore not enough to explain or guarantee improvement.

1.5 The inverse direction survives a controlled change of shape

The attenuation under selection does not mean that the inverse relation is confined to comparisons of unrelated bodies. Keeping polygon normals fixed and setting both factors' support heights to one gives opposite changes in R and the numerical systolic ratio in fifteen of sixteen pairs. In ten pairs R decreased and the ratio increased; in five the reverse happened. Both quantities increased in the remaining pair. Thus the sign relation persists for most of these controlled changes, even though making the factors tangential does not always raise the ratio.

Separate normalization of the factors preserves the ratio: for $a, b > 0$,

$$\text{diag}(aI_2, bI_2) = \sqrt{ab} \text{diag}(\sqrt{a/b} I_2, \sqrt{b/a} I_2)$$

is a dilation followed by a symplectic map. Thus replacing unequal support heights is the geometric change being tested.

The evidence is stronger for direction than for size of effect. The Spearman correlation between the two changes is -0.55 overall and only -0.095 in the 4×6 group. All tested ratios are at most 0.742, so this is not a deformation test of the high-ratio tail. It is consistent with shared geometric variation producing the bulk association, but does not isolate a causal role for ridge area or explain the whole pattern. The paired constructions and checks are in Appendix A.5.

1.6 Covariance balance gives a second geometric signal

Ridge area is not the only summary associated with the ratio. In an exploratory analysis, more nearly equal symplectic covariance scales of the vertex set were associated with larger ratios. For vertices $v_1, \dots, v_N$, define

$$\bar{v} = \frac{1}{N} \sum_i v_i, \quad \Gamma = \frac{1}{N} \sum_i (v_i - \bar{v})(v_i - \bar{v})^T.$$

In Williamson normal form, Γ has diagonal $(\nu_1, \nu_2, \nu_1, \nu_2)$, with $0 < \nu_1 \leq \nu_2$. The ratio $\rho = \nu_2/\nu_1$ measures the imbalance between the two symplectic covariance scales.

Like R , this score is invariant under affine symplectic maps and common dilation. Centering removes translations. For symplectic S , $\Gamma(SK) = S\Gamma(K)S^T$, so with $J = \begin{pmatrix} 0 & I_2 \\ -I_2 & 0 \end{pmatrix}$,

$$J\Gamma(SK) = S^{-T}(J\Gamma(K))S^T.$$

The eigenvalues $\pm i\nu_1, \pm i\nu_2$ are unchanged. Dilation multiplies both ν_j by the same square factor. For a product with planar vertex covariances A, B , moreover, $\Gamma = \text{diag}(A, B)$ and $(J\Gamma)^2 = \text{diag}(-BA, -AB)$; thus ν_1^2, ν_2^2 are the eigenvalues of AB .

Low ρ selected larger mean ratios in every side-count group in each of two fresh pools. The twenty gains over comparison bodies ranged from 0.260 to 0.440, with mean 0.333. Against the combined low- R , low- C selector, its mean advantage was only 0.014, and the differences changed sign across groups. Both geometric summaries therefore support selection in the tested populations; neither was uniformly better there.

Their usefulness also survived one change of sampling law. For tangential factors obtained by setting the support heights to one, the mean gains were 0.259 for covariance selection and 0.225 for ridge selection, averaged equally over the 4×6 and 6×6 groups. This is one conditioned source, requiring both original and changed polygons to be valid. It establishes a particular transfer, not independence from the sampling distribution. Details are in Appendix A.4.

1.7 Balancing the factors improves most tested products

Removing the planar vertex-covariance anisotropy of both factors raises the mean numerical ratio from 0.342 to 0.654 in a matched comparison of 320 products, with increases in 314 and decreases in six. At the same time, the pooled within-group rank correlation between R and the ratio weakens from -0.929 to -0.249 . Thus affine shape variation of the factors contributes materially to the broad association in this sample.

The transformation has a precise geometric meaning. For a factor with vertex covariance A , set

$$L_A = (\det A)^{1/4} A^{-1/2}.$$

Then $\det L_A = 1$ and $L_A A L_A^T = \sqrt{\det A} I_2$. Applying L_A and L_B to the two factors preserves their areas and combinatorics and makes both vertex covariances scalar matrices. The product therefore has $\rho = 1$. These independent factor maps need not combine to a symplectic map, so capacity can change even though volume is preserved.

The mean ridge sum falls from 23.739 to 11.652, decreasing in 306 of the 320 products. The remaining relation depends on the side counts: the balanced 3×3 group has positive rank correlation, while the other nine groups retain negative correlations. The transformation changes several geometric properties together. It therefore identifies a useful shape deformation, not covariance imbalance as the sole cause of the bulk association. Nor do its six decreases permit a universal improvement rule. The target-blind choice of original bodies, complete paired evaluations and numerical checks are in Appendix A.6.

1.8 What could explain the bulk association?

The controlled changes give substance to one possible explanation of the bulk pattern. Many original products leave a large improvement available under factor balancing, and removing that shape variation weakens the association. Common perturbations of relatively inefficient shapes could therefore account for much of the broad inverse ordering, while other features govern differences among the better bodies. The height changes supply a second concrete deformation with a mostly inverse direction. Neither experiment measures geometric distance to a local optimum or establishes an additive-noise model.

A different explanation is an underlying relation involving a geometric quantity that R measures only imperfectly. Extreme R can then reflect both favorable geometry and deviations specific to the proxy. Here noise would mean differences between bodies not captured by the relation, not numerical error in a fixed-body evaluation. The residual associations after balancing leave room for such a relation; they do not identify it.

These explanations are compatible, and the results do not choose between them. The observed attenuation is stronger than in the fitted Gaussian reference, but other dependence structures can behave differently under selection. Neither convenient account need describe the full phenomenon. The exact pentagon identity, the global counterexample and the controlled shape changes provide constraints that a broader explanation must respect.

1.9 Using evaluated ratios to guide search

A weak explanation of the ridge signal need not prevent practical search. Adapting the product sampling law to evaluated ratios improved both the maximum and the quality of several leading bodies in three paired runs. For 5×5 products, the cross-entropy method [Boe+05] shifts a fitted distribution towards the quarter of the current population with largest ratios, then samples again. Under equal evaluation counts, the results were:

Seed	Maximum		Top-eight median	
	IID	Adaptive	IID	Adaptive
1	.760	.799	.729	.782
2	.811	.831	.754	.776
3	.760	.856	.722	.814

The top-eight statistic is the median of the eight highest ratios after removing repeated exact dual-vertex arrays. Both arms share their initial population. The fitted coordinates describe log ratios of normal-angle gaps, centered logarithmic dual-normal lengths and a relative angle; the latter uses a circular mean update. The coordinate-dependent evaluator admission condition acts differently on the original sampling coordinates and the adaptive normalized representation. The observed advantage therefore concerns this implemented combination of adaptation and admission. Appendix A.9 records the update, budgets and admission history.

The prescribed ambient rotations $K \mapsto UK$, $U \in SO(4)$, also lost to fresh sampling under equal evaluation budgets. These limited comparisons are recorded in Appendix A.5.

1.10 Competing branches in local search

The local-search comparison favors methods that account for several competing branches of the capacity calculation. A single selected branch can point uphill while the minimum of the branches decreases. For example, $\min\{x, -x\}$ decreases in either direction from zero although each individual branch has an increasing direction. The corresponding search question is whether retaining nearby or previously active branches helps the numerical systolic ratio increase.

Seven fixed implementations started from the same 64 generic ten-facet bodies, under a common cap of 128 full evaluations and a nominal one-second accumulated evaluator-plus-optimizer budget. The implementations retaining branches from earlier points attained the largest median terminal values:

Implementation	Median best value	Across-start 10–90% range
Four-anchor history	.985	.957–.999
History with transition prediction	.982	.956–.994
Finite-gap model	.977	.948–.995
Literal gradient	.881	.790–.924
Safeguarded gradient	.855	.780–.916
CMA-ES	.818	.718–.869
Coordinate poll	.601	.358–.787

The finite-gap method models branches near the current minimum. The two history methods also retain earlier branches; one predicts additional branches from transitions. CMA-ES [Han16]

and coordinate polling supply population and direct-search comparisons. These are rankings of the fixed implementations under this allocation, not of optimally tuned algorithm families. The historical numerical objective uses floating-point geometry and volume, and not every original executable or proposal trace was recovered. Appendix A gives the retained steps, provenance and budget interpretation. The ranges in the table show variation across starting bodies; all 448 terminal values are below one.

1.11 A failed probe does not establish a local maximum

The endpoint diagnostics show that a high terminal ratio need not mean local convergence. One optimizer endpoint decreased along every tested signed basis direction, yet a continuation increased its ratio by 0.00223 along oblique moves of total normalized path length 0.01. All three selected endpoints in that continuation accepted ten improving steps, while HKO accepted none. HKO’s local status is established by its theorem; the finite test supplies a consistent control observation.

A regular-pentagon control on a known increasing rotation path illustrates the same problem independently. In a suite of basis, random and prescribed rotation probes, its only positive changes were the twelve rotation probes; the generic directions missed the available increase. The regular triangle–hexagon product, rotated square–square product and CH2021 example had no nominal gain above 10^{-12} in that suite. Their misses do not exclude untested directions or changes in facet count. The probe design and numerical limitations are in Appendix A.8.

The chapter therefore supports a strong coarse association, an exact law on one rotation family, and useful finite search improvements. It does not supply a universal capacity law or a general route to local optimality. None of the random samples or search comparisons produced a numerical ratio above one. For products with a triangle or quadrilateral factor, the Viterbo inequality is already known [BMP26, Theorem 1.2 and Section 4]; such products can illuminate the pattern but cannot yield counterexamples while remaining in that class. The remaining product families and generic polytopes leave open both questions: what geometric relation explains the signal, and what search directions remain effective near large ratios?

A Data-science methods and additional results

This appendix records the population, prediction, selection and search methods supporting Section 1, together with the additional numerical diagnostics and their interpretation limits.

A.1 Sampling and retained data

A polytope was specified by inequalities

$$K = \{x : \langle n_i, x \rangle \leq h_i, \quad i = 1, \dots, F\}.$$

For the generic family, each n_i was independent and uniform on the unit sphere in $\mathbb{R}^4$: we normalized a vector of four independent standard Gaussian coordinates. The heights were independent and uniform on $[0.8, 1.2)$. A proposed body was retained only if it was bounded and every proposed inequality defined a facet. The construction also rejected zero or near-duplicate normals. These conditions are part of the sampling law; “random polytope” here does not mean a uniform choice from all convex bodies.

The original population contains 512 bodies for each generic facet count and 1,024 for each product side-count pair. The retained generator records identify seed 42; product seed/attempt metadata cannot be independently reconstructed from the recovered cache. The source geometries, row identifiers and stored numerical values are nevertheless matched for every body. The reproduction records are listed in Table 2.

A.2 Numerical evaluation and retrospective checks

The complete historical source geometry was recovered and matched to all 14,336 table rows. A geometry-only rebuild restored six later descriptors; all 645,120 overlapping table cells agreed. This retained the historical volume normalization and did not evaluate capacity.

A separate current-evaluator run attempted every body. It returned 14,245 successful evaluations, 54 input-policy failures and 37 scalar-tolerance failures. Among successful rows, the largest relative differences were $1.06 \cdot 10^{-14}$ in capacity, $3.14 \cdot 10^{-14}$ in volume and $3.23 \cdot 10^{-14}$ in the ratio. No row is dropped from the original statistical analysis. The original failures are retained alongside the separate follow-up below.

A subsequent interval-preserving evaluation recovered results for 90 of those refusals. The 37 scalar-tolerance refusals yielded capacity intervals without changing their input bodies. Exact power-of-two rescaling brought 53 of the 54 norm-policy failures into the admitted range; fresh geometry checks preceded evaluation, and homogeneity transferred their interval endpoints back to the original bodies. Across both runs, 14,289 bodies have accepted capacity scalars at tolerance 10^{-10} and another 46 have wider capacity intervals. One body, `random_F8_s3_45`, remains unresolved. No policy or tolerance was relaxed. The complete sample, including this body, remains in the historical statistical analysis.

The current evaluator reconstructs the geometry of the stored binary64 input and returns rational product capacities or outward capacity bounds. Its volume calculation remains floating-point. Agreement of central values does not mean that a narrow new interval contains an independently rounded old value: 5,583 old capacity scalars lie outside the new intervals despite the small differences above. The initial reevaluation corroborates the numerical values on its successful bodies; it does not certify the original execution or the complete ratio. Prospective selection experiments and optimizer traces have their own evidence records and are not validated by this baseline comparison.

A.3 Prediction models and feature families

Prediction experiments gave another view of the signal. A random forest [Bre01], which averages predictions from many decision trees, used 39 descriptors: 27 describing face counts and incidences, and 12 summarizing ridge areas. The test split held out groups defined by capacity-source label and facet count, leaving 8,192 training rows and 6,144 test rows. Its test coefficient of determination was $R^2 = 0.885$. Here

$$R^2 = 1 - \frac{\sum_{i \in \text{test}} (s_i - \hat{s}_i)^2}{\sum_{i \in \text{test}} (s_i - \bar{s}_{\text{test}})^2}$$

compares squared prediction error with prediction by the test-set mean. Here s_i is the observed numerical systolic ratio and $\hat{s}_i$ is the model prediction. The mean absolute error was 0.0495. This was one group split, not repeated cross-validation across independent datasets.

A separate feature-family comparison used histogram-based gradient-boosted regression [Sci], which successively fits trees to improve the prediction. Under its common training/test split, ridge-area features alone gave $R^2 = 0.887$, whereas combinatorial counts alone gave 0.043. The comparison points to information in the ridge areas beyond those counts. Its split and feature list differ from the random-forest experiment, so the two scores do not rank the learning algorithms against each other.

A.4 Selection and comparison rules

The 100,000-product concentration experiment specified a positive aggregate mean difference and positive differences in at least seven of ten groups as its continuation criterion. It found

eight positive groups and a mean difference of about 0.034. This is the descriptive design criterion; no significance claim is attached to passing it.

For the tangential-generator comparison, controls were selected by a fixed hash order, before capacity evaluation, from the eligible pool after removing the union of the ridge and covariance selections. They are unselected same-generator comparisons, not an asserted random sample of the full pool.

The opposing 3×6 and 4×4 rotation paths contain four recorded positions each. Their simultaneous decrease in ridge score and numerical ratio is a finite observation; monotonicity at every intermediate angle was not established.

Initial ridge selection. The preliminary pool contained 100,000 products. Thirty scalar selection sets and their baselines required 1,675 distinct evaluations; their union had maximum recorded ratio 0.868. The low- R rule selecting ten bodies per side-count group gave mean 0.6261166821, against 0.3155039434 for ten comparison bodies from each group. Comparison bodies were selected by a rule-specific fixed hash order excluding the selected set. Overlapping rules are not independent repetitions.

Restriction to the low-ridge tail. The conditional comparison uses 1,000 evaluated products from the lowest one percent by R , with 100 in each side-count group. Spearman correlations within groups range from -0.315 to $+0.301$; pooling ranks computed separately within each group gives $+0.03197$. Selecting the most extreme tenth increases the mean in only two of ten groups. The broad and conditional panels are different generated samples, so their comparison concerns a change in population and score range. The retained analysis is in `tail-dependence-feasibility/artifacts/current/`, under the methods directory in Table 2.

Conditioning the original dataset. A separate analysis retains the lowest- R or highest-ratio fraction within each of the eighteen original sample groups. At retained fractions 100%, 20%, 10%, 5%, 1%, both arms contain respectively 14,336, 2,874, 1,446, 728 and 158 bodies. There are no ties across these cutoffs. Each retained group is reranked independently; its two rank columns are divided by the group’s retained size, then pooled for a Spearman correlation. Thus the summary weights bodies rather than giving each group equal weight. The low- R values are $-0.944, -0.434, -0.222, -0.113, +0.035$, and the high-ratio values are $-0.944, -0.496, -0.286, -0.239, -0.036$. The smallest panels contain only six generic or eleven product bodies per group. These are descriptive statistics under conditioning, not evidence of independence, systematic reversal, or the effect of a shape intervention. Row identities, cutoffs and a second rank calculation were independently checked; the reproducible packet is `experiments/sys-datascience/methods/selection-mechanism/`.

Gaussian rank reference. Each original sample group is assigned a bivariate Gaussian reference with Pearson correlation $2 \sin(\pi r_s/6)$, where r_s is its observed full-sample Spearman correlation. One thousand simulations preserve all eighteen group sizes and apply the same conditional ranking statistic. At the five-percent cutoff, the reference means are -0.695 for both selection directions. The central simulation intervals are $[-0.738, -0.649]$ for high-ratio selection and $[-0.736, -0.647]$ for low- R selection, against observed values -0.239 and -0.113 . These are pointwise simulation references with fitted parameters held fixed. The model was considered after seeing the attenuation; this is not a prespecified model test or a confidence interval for the observed correlation. The reference, reproducible simulation and input hashes are in the same packet’s `gaussian-result.md` and `gaussian-artifacts/`.

Covariance selection. Two fresh pools of 50,000 candidates compared the lowest 0.5% by ρ , the lowest 1% by R followed by its lower half by C , and 25 comparison bodies per side-count group. An ordinary covariance conditioning cutoff excluded three candidates across the two pools. Controls were selected by fixed hash order from the eligible bodies after removing both geometric selections. There were 500 memberships in each selection and 1,436 distinct evaluations. All twenty covariance-minus- control group effects were positive, ranging from 0.259639 to 0.440029. The mean effect was 0.333123, while the mean covariance-minus-ridge effect was 0.014488; thirteen of the latter twenty effects were positive.

The retained analysis also gives nominal Student- t intervals. They are $[0.310, 0.356]$ for covariance versus controls and $[-0.0149, 0.0439]$ for covariance versus ridge selection, using the twenty group effects and nineteen degrees of freedom. The groups have fixed, different side counts, and no common-mean sampling model or calibrated coverage for repeating the complete selection procedure was established. The main text therefore reports the observed consistency across groups.

Generic and tangential transfer. The generic ten-facet experiment selected 100 of 10,000 bodies by lowest mean normalized ridge area, with 100 disjoint, deterministically selected comparison bodies. The mean contrast was 0.287, with a percentile interval $[0.250, 0.324]$ from 20,000 resamples. The ten most extreme scores minus the other ninety had mean contrast -0.0318 and interval $[-0.0946, 0.0331]$. These resamples hold the selected panels fixed; they do not regenerate the pool, rerank its bodies or reselect its tail. The selected-panel rank correlation between mean ridge area and ratio was $+0.149$.

The tangential-source comparison required both original and changed factors to be admissible and normalized the changed factors separately to area one. One pool supplied 3,200 eligible bodies in each of the 4×6 and 6×6 groups. Each selection had sixteen memberships per group. Five overlapping selections gave 91 distinct evaluations. The equally weighted group effects were 0.259 for covariance versus controls and 0.225 for ridge versus controls; their within-panel resampling intervals were $[0.192, 0.329]$ and $[0.160, 0.296]$. These condition on the observed panels. Shared controls make the two comparisons dependent.

A.5 Paired height changes and ambient rotations

The paired support-height experiment used sixteen independently generated pairs of polygons, eight 4×4 and eight 4×6 . For each, the original product was compared with equalizing heights in the first factor, the second, or both; all four constructions had to be valid before any ratio was evaluated. Each factor was scaled to area one. All 64 evaluations succeeded. Changing both factors improved eleven pairs and worsened five, with mean numerical change 0.0153. The group means were 0.0040 and 0.0266 respectively. This is the jointly admissible source, not an unconditioned statement about all polygon pairs.

Recomputing R from the retained paired geometries gives opposite signs of ΔR and $\Delta \widehat{\text{sys}}$ in fifteen of sixteen both-factor changes. The one exception has R increasing from 11.074 to 11.218 while the numerical ratio increases from 0.419 to 0.436. The one-factor comparisons have opposite signs in thirteen and fourteen of sixteen pairs. All 64 bodies have $R \geq 11.065$ and numerical ratio at most 0.742. These results reuse the retained capacity evaluations; no new capacity calls enter the comparison. Reconstruction, row identity and hash checks are recorded in `experiments/sys-datascience/methods/selection-mechanism/controlled-result.md`.

For ambient rotations, the action is

$$P \times Q \mapsto U(P \times Q), \quad U \in SO(4).$$

It preserves Euclidean shape and volume. A nonsymplectic U can change capacity, and the image need no longer be a Lagrangian product for the fixed symplectic form. One allocation

evaluated a source product, a symplectic rotation control and fourteen prescribed orthogonal rotations; the other evaluated sixteen fresh products, sharing the source body. There were four comparisons, two each for 3×3 and 4×4 . The schedule used fourteen fixed matrices, not uniform sampling over $SO(4)$. The shared starts gave 124 distinct requests and 128 memberships; symplectic controls preserved numerical ratios within $4 \cdot 10^{-16}$.

Rotation improved three starts, but fresh sampling attained a larger maximum in all four comparisons. In one quadrilateral example, rotation raised 0.328 to 0.529, while fresh sampling reached 0.774. Rotation evaluations also took longer. These are observations about the prescribed schedules and starts, not a comparison of all ambient rotations with all products from the source law.

A.6 Factor covariance balancing

The paired comparison uses 32 original products from each of the ten side-count groups. The selection is fixed by hashes of row identifiers, without using the target ratios. For planar vertex covariances A, B , the maps $(\det A)^{1/4}A^{-1/2}$ and $(\det B)^{1/4}B^{-1/2}$ have determinant one and make each covariance scalar. There is one original and one transformed body for every selected row, evaluated through the same current product-capacity route. All 640 evaluations succeeded.

The mean numerical ratio changes from 0.341888 to 0.653683; 314 paired changes are positive and six are negative. The mean R changes from 23.738833 to 11.652347, with 306 decreases and fourteen increases. The pooled within-group rank correlation changes from -0.928922 to -0.248680 . The balanced triangle-product group has correlation $+1$; the other nine groups range from -0.571 to -0.098 . Thus the pooled weakening does not mean that every transformed group has negligible or negative dependence.

The transformed covariance ratios differ from one by at most $5.1 \cdot 10^{-13}$, and relative factor-area changes are below $3.2 \cdot 10^{-13}$. The original current evaluations agree with their retained historical numerical ratios within $1.3 \cdot 10^{-15}$. These are paired transformations of retained source bodies, not a new independent population or a test isolating the scalar ρ from other changed geometric properties. The producer, frozen inputs, evaluator identity and complete paired results are retained in `experiments/sys-datascience/methods/selection-mechanism/balance-artifacts/`.

A.7 Finite-budget local optimizer comparison

The run configuration labels the 64 starting bodies as held out. The saved hashes identify the seven configurations used, but do not establish that these bodies were absent from earlier development or tuning. A later development configuration reuses 16 starts. Their earlier selection history is therefore unresolved; this does not show that the reported outcomes were used to select the configurations.

Every run used the same allocation rule: a nominal 1000 ms start-new-work cutoff for charged full-evaluator plus optimizer time and a cap of 128 full historical-evaluator calls. In-progress work could complete atomically past the cutoff, and methods could terminate early, so realized compute differs across runs. The 448 traces contain 28824 full calls. Using the historical numerical objective described in Section 1.10, four-anchor branch history has the largest comparison-block median best value, 0.9848, followed by transition prediction at 0.9816 and the finite-gap affine model at 0.9772. No run reaches a recorded numerical ratio of one.

The implemented optimizer steps. Except for the literal gradient, the methods remove the fifteen infinitesimal symmetry directions from the forty dual-normal coordinates. The local models and coordinate poll recompute this transverse space at their anchors; CMA-ES keeps its initial space. The literal gradient uses the ambient coordinates. Every outer proposal is checked by a fresh geometry reconstruction and the configured full orbit-word search. The

implementations below are frozen comparison points; their parameters are not claimed to optimize the corresponding algorithm families.

Table 1: One outer iteration of each optimizer in the recorded comparison.

method	implemented step
literal and safeguarded branch gradients	Differentiate one currently minimizing branch. The literal method takes $a^+ = a + 0.01D_a s_\sigma(a)$ even after a decrease. The safeguarded method projects and normalizes this gradient, accepts only a fully evaluated improvement, and expands or halves its distance after acceptance or rejection.
finite-gap affine model	Keep the current branch gaps in a 10% action window, linearize their values, and maximize the minimum affine value on a local ball. Fully evaluate the proposal and use the same expand-or-halve distance rule.
four-anchor branch history	Union candidate words from the four most recent full searches, using a 30% action window and normalized beta allowance 0.3. At a nearby point, re-solve the named KKT branches, discard transition-infeasible and beta-inadmissible solutions, linearize the remaining values, and maximize their affine minimum. Take at most two internal moves before the next full evaluation.
history with transition prediction	Use the preceding method, but for a normalized internal move of length at least 0.08, also predict newly feasible transitions in the move direction and evaluate the words thereby generated.
CMA-ES	Fully evaluate a population of eight Gaussian perturbations, then update the mean, covariance, and normalized scale from their ranking.
coordinate direct search	Poll both signs of the 25 transverse basis axes. Move to the best improvement and double the radius, or halve the radius if no poll point improves.

The two history methods accept any usable outer evaluation, including a decrease; their reported best-so-far value therefore differs from their current iterate. The safeguarded gradient and finite-gap method accept only improvements, multiplying the relative step length by 1.25 on success and by 0.5 on failure. The history methods use up to two internal model steps between full evaluations.

The companion endpoint packet does not test local maximality. Signed polls along the transverse basis find an improving direction at seven of 16 one-second branch-history endpoints and two of the corresponding five-second endpoints. Separate validated continuation paths show that even an endpoint with no improving basis direction can retain appreciable ascent between the tested axes. These observations diagnose insufficient finite budgets; they do not estimate the distance to a local maximum.

Detailed HKO calibration. On 16 known perturbations along four development directions from HKO, the multi-branch continuation accepted no move at HKO and improved every perturbation. One step recovered 93.4%–100% of the known evaluator gap and removed 77.9%–100% of the distance to HKO. In contrast, the gradient of one current minimizing branch recovered −22.1%–41.4% of the gap.

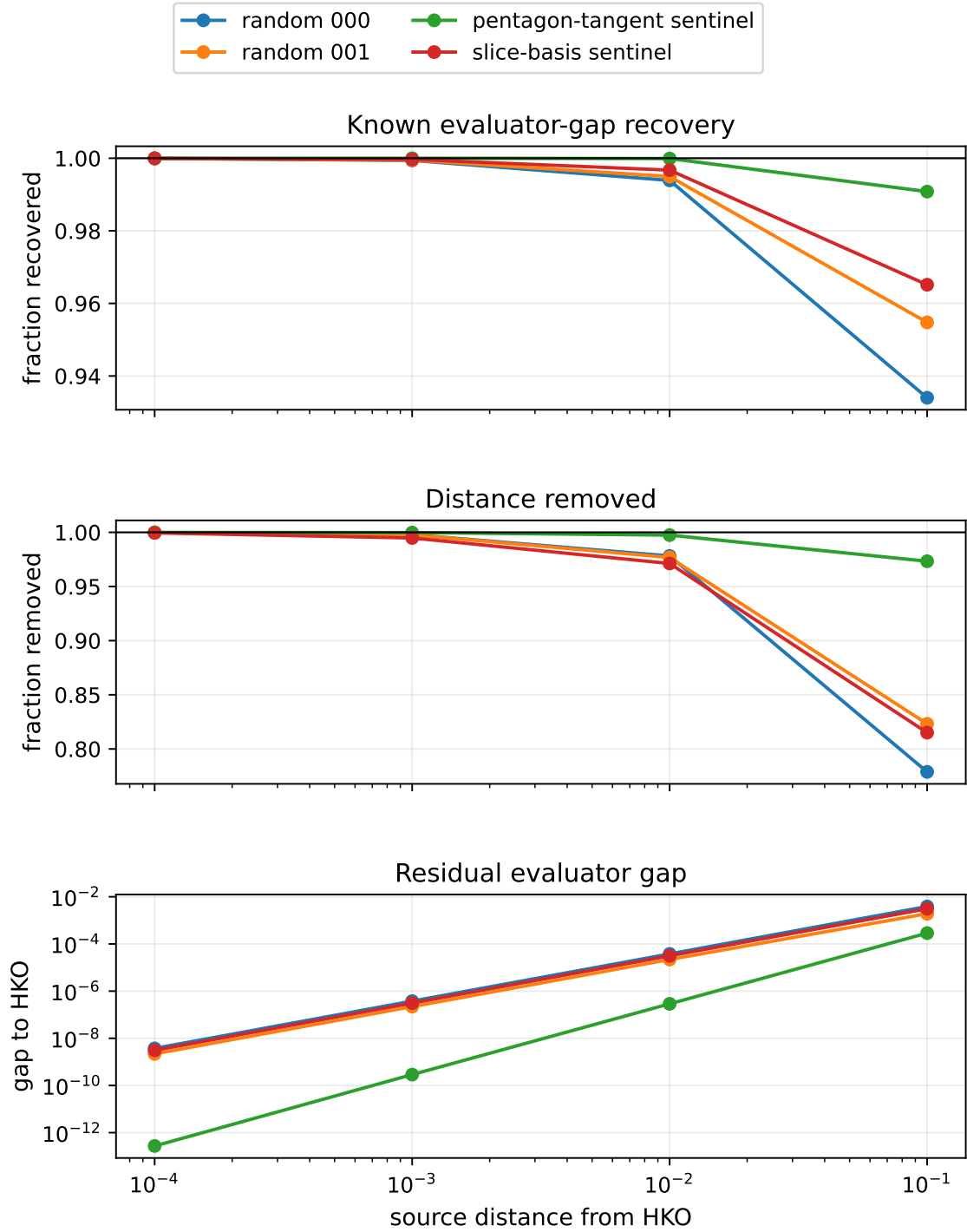


Figure 4: One-step calibration on known perturbations of HKO. From top to bottom, the panels show recovered evaluator gap, removed distance, and residual evaluator gap.

Selected finite-step failure audit. At the eight highest four-anchor tuning endpoints, five admitted one validated improvement, by $1.57 \cdot 10^{-6}$ to $4.61 \cdot 10^{-5}$. Across the 80 multi-branch proposals, 52 decreased the recomputed evaluator value despite a represented target winner and positive affine prediction; 40 of these losses retained determinate, unchanged geometry. All 40 current-branch gradient proposals decreased.

The three-proposal audit compared two such failures with one positive control. At normalized radius 10^{-5} , the failures perturbed their named KKT matrices by 218 and 67 times the smallest base eigenvalue magnitude and crossed zero; the control had ratio 1.53, did not cross zero, and kept the correct sign. At the smallest tested radius, 10^{-8} , central differences approximated the implemented analytic derivatives with relative errors below 1% for the named action and 1.9% for the named branch ratio. Binary64 and exact-arithmetic geometry reconstruction agreed at all 39 audited points, and relative volume discrepancies were below 10^{-15} . For these selected cases the observed failure is finite-distance nonlinearity of an ill-conditioned KKT branch.

These diagnostics use selected controls rather than population samples. They do not provide a convergence theorem, a population error rate, or a universal conditioning cutoff.

A.8 Endpoint probes and continuation

Sixteen history-method endpoints were chosen across the observed terminal performance range. Each received both signs of 25 transverse basis directions at relative radii 10^{-3} , 10^{-4} and 10^{-5} , giving 150 probes per endpoint. Seven endpoints had a raw positive change and nine had none. Rank-based endpoint selection prevents interpreting these fractions as a population convergence probability.

After a larger budget on those starts, continuation examined three selected endpoints and HKO. It tested branch-model and selected-gradient proposals at five decreasing step lengths, with fallback signed basis polling, and allowed ten accepted improvements. All three endpoints took ten improving steps; HKO took none. One endpoint with no improving signed basis direction gained 0.00223 along oblique moves of total normalized path length 0.01.

The separate triangle-hexagon, square-square, CH2021 and pentagon-control suite used basis, projected random and prescribed rotation directions at three radii. All 1,482 probes were valid. The three targets had no nominal gain above 10^{-12} ; the pentagon control had twelve, all from rotations. Incomplete historical source provenance and broad capacity-derived bounds prevent interpreting the nominal misses as certified local maximality.

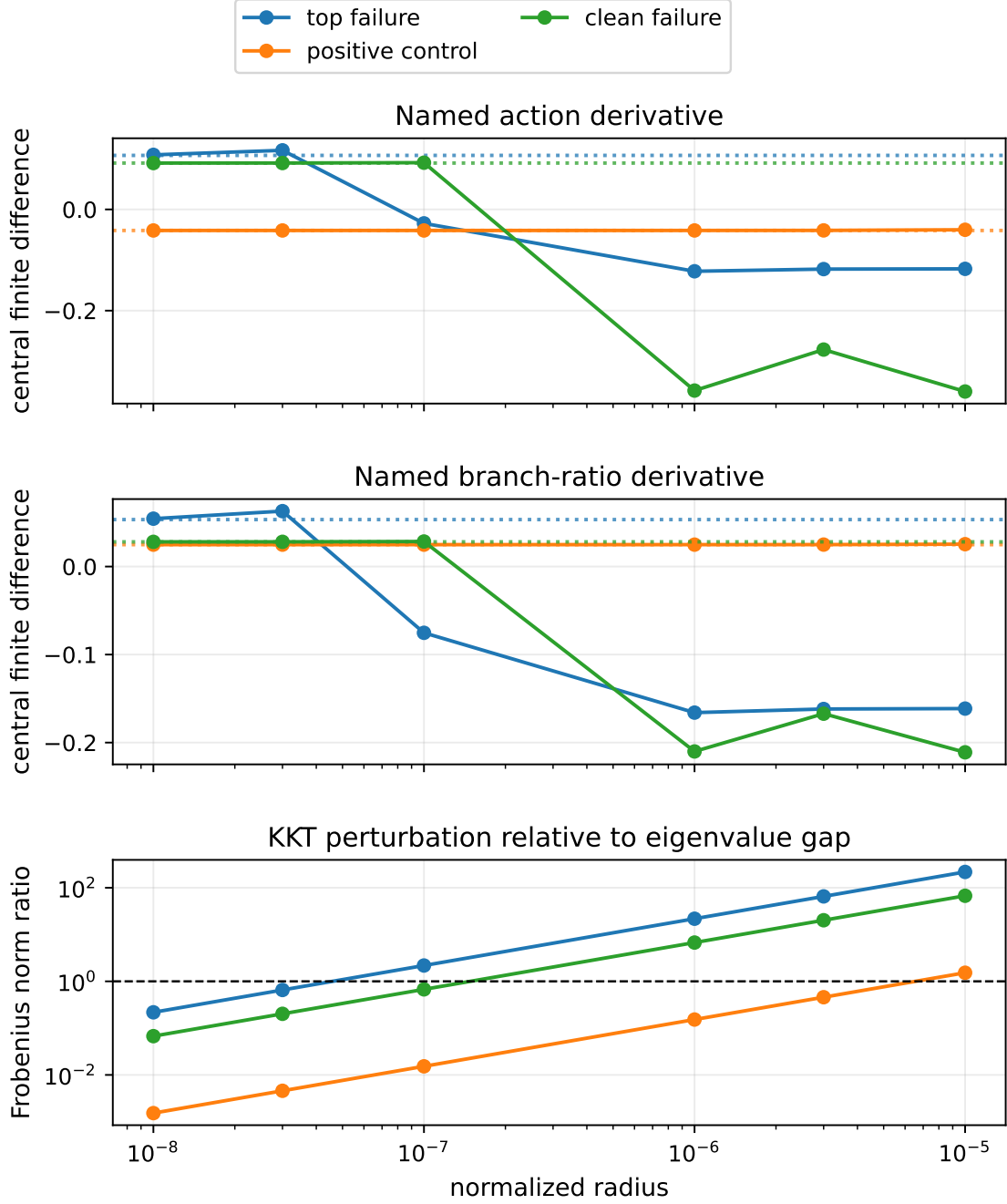


Figure 5: Finite differences for two failed affine proposals and one positive control. Prediction changes regime when the KKT perturbation becomes large relative to the smallest source eigenvalue magnitude (bottom). Dotted lines in the top two panels are the analytic derivatives.

A.9 Adaptive product sampling

Both arms began with the same 64 independently generated products. Four populations including initialization gave 256 charged evaluations per arm. The adaptive update used the sixteen bodies with largest ratios. Across three runs, shared initial populations gave 1,344 actual evaluations and 1,536 arm memberships; all actual evaluations succeeded and all recorded ratios were below one. The top-eight statistic removes repeated exact dual-vertex arrays, retaining different arrays with equal scores.

For reproducibility, let $g_1, \dots, g_5 > 0$ be the cyclic normal-angle gaps of a factor and let r_i be its dual-normal lengths. Set

$$\ell_i = \log(g_i/g_5) \quad (i = 1, \dots, 4), \quad z_i = \log r_i - \frac{1}{5} \sum_{j=1}^5 \log r_j.$$

We store the first four z_i and recover $z_5 = -\sum_{i=1}^4 z_i$. Reconstruction uses

$$g_i = \frac{2\pi e^{\ell_i}}{1 + \sum_{j=1}^4 e^{\ell_j}}, \quad g_5 = \frac{2\pi}{1 + \sum_{j=1}^4 e^{\ell_j}}, \quad h_i = e^{-z_i}.$$

The first normal angle is zero in the first factor and ϕ in the second. The initial IID encoding chooses a cyclic origin by lexicographically minimizing the sequence of gap/length pairs. Sampled adaptive coordinates are retained for the updates without another cyclic relabelling.

For an ordinary coordinate j , write $\hat{\mu}_{E,j}$ and $\hat{v}_{E,j}$ for the mean and population variance of the sixteen elites. The update is

$$\mu_j^+ = \frac{1}{2}\mu_j + \frac{1}{2}\hat{\mu}_{E,j}, \quad v_j^+ = \max\{\frac{1}{2}v_j + \frac{1}{2}\hat{v}_{E,j}, 0.05v_{0,j}\}.$$

Initialization uses the population moments of the shared 64 bodies. Coordinates are drawn independently with these Gaussian moments. The phase uses circular means, a unit-vector blend of old and elite means, and variance of angular deviations wrapped into $[-\pi, \pi)$. Samples are wrapped modulo 2π . A vanishing resultant retains the previous mean; initial fallback uses the smallest-ID candidate's phase. Ties in elite selection are broken by candidate ID.

Each population permits at most 640 construction attempts for 64 accepted bodies. Failed geometry and unsupported evaluator inputs are rejected before target evaluation; an incomplete population does not trigger an update. The evaluator admission includes finite, nonredundant bounded geometry and primal and dual infinity-norm bounds between 10^{-3} and 10^3 . The three seeds are 202609140301, 202609140302 and 202609140303. The retained report records the stopped first version and the subsequent supported version separately.

The source restriction needs particular care. An initial version stopped when one body exceeded the evaluator's admitted primal-vertex infinity-norm range. The completed version applied the evaluator's existing admission conditions to both proposal arms before accepting a body for evaluation. This was a procedural revision after seeing the failure. Both arms reject unsupported inputs, but the admission test acts on the actual coordinate representation: IID bodies retain their original representation and adaptive bodies use the normalized reconstruction. It is not a symmetry-invariant condition on all pentagon products. The result is therefore for this implemented, admission-conditioned comparison.

The complete run took 215.8 seconds: 182.2 seconds for construction and admission checks, 32.9 seconds in evaluator processes, and 0.053 seconds for distribution updates. Including construction and charging initialization to each arm, the adaptive runs used about 35–36 seconds per seed and IID about 46–51 seconds. Equal evaluation counts did not imply equal cost. These measured costs favor the adaptive arm in this run; they are not the outcome of a separate equal-wall-time experiment.

A.10 Evidence and reproduction index

Paths in Table 2 are relative to the accompanying repository. Each packet identifies its inputs and analysis; the original historical optimizer executable and full proposal traces were not recovered.

Table 2: Source packets for the chapter.

Result	Repository source
Historical population and replay	docs/ds-evidence-closure/README.md;
Restored features and reevaluation	experiments/polytope-datasets/retained-lineage.md docs/ds-retrospective-revalidation/geometry-full/README.md; docs/ds-retrospective-revalidation/target-full/README.md; docs/ds-retrospective-revalidation/repair/README.md
Feature models	experiments/sys-datascience/methods/prediction-ranking/; experiments/sys-datascience/methods/standard-baseline-p2/
Fresh selection	experiments/sys-datascience/methods/extreme-scalar-rejection-proposer/; experiments/sys-datascience/methods/extreme-scalar-rejection-proposer/artifacts/covariance-rho-frozen-validation/
Transfer	experiments/sys-datascience/methods/alternative-source-transfer/; experiments/sys-datascience/methods/generic-ridge-tail-stage1-target/
Paired support heights	experiments/sys-datascience/methods/paired-tangentialization/
Orientation allocation	experiments/sys-datascience/methods/orientation-allocation/
Adaptive product sampling	experiments/sys-datascience/methods/diagonal-cem-pilot/
Local optimization and diagnostics	experiments/dev-gradient-ascent/optimizer-comparison/; docs/ds-evidence-closure/search-account.md
Conditional and controlled ridge analyses	experiments/sys-datascience/methods/selection-mechanism/
Conditional ridge correlations	experiments/sys-datascience/methods/tail-dependence-feasibility/
Product ridge geometry	docs/ds-evidence-closure/ridge-mathematics.md; experiments/sys-datascience/methods/ridge-endpoint-path/

References

- [BMP26] Alexey Balitskiy, Ivan Mitrofanov, and Alexander Polyanskii. *Triangle covering problems and the Viterbo inequality in the plane*. Version 1, Theorem 1.2 and Section 4. 2026. arXiv: 2603.12495. URL: <https://arxiv.org/abs/2603.12495v1> (visited on 09/18/2026).

- [Boe+05] Pieter-Tjerk de Boer et al. “A Tutorial on the Cross-Entropy Method”. In: *Annals of Operations Research* 134 (2005), pp. 19–67. URL: <https://people.smp.uq.edu.au/DirkKroese/ps/aortut.pdf>.
- [Bre01] Leo Breiman. “Random Forests”. In: *Machine Learning* 45 (2001), pp. 5–32. DOI: 10.1023/A:1010933404324.
- [Gar02] Richard J. Gardner. “The Brunn–Minkowski inequality”. In: *Bulletin of the American Mathematical Society* 39 (2002), pp. 355–405. URL: <https://www.ceremade.dauphine.fr/~carlier/Brunn-Minkowski>.
- [Han16] Nikolaus Hansen. *The CMA Evolution Strategy: A Tutorial*. 2016. URL: <https://arxiv.org/abs/1604.00772>.
- [Sci] Scikit-learn developers. *Histogram-Based Gradient Boosting*. URL: <https://scikit-learn.org/stable/modules/ensemble.html#histogram-based-gradient-boosting> (visited on 09/14/2026).